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Tunnel performance prediction through degradation inspection and digital twin construction

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ABSTRACT

Civil infrastructures inevitably undergo a degradation process due to extended periods of operation, frequent service requests, and weathering. This paper proposes a methodology for predicting necessary interventions and presents a database of inspection information for tunnels. The model developed correlates performance loss, measured by the tunnel serviceability index (*TSI*), to defects detected through a 2D grid composed of classified unit cells converted into a virtual representation. To support the framework developed, intervention time estimates were compared to stochastic degradation models presented in the literature. A limit analysis was performed using inverse exponential equations, utilizing data from different sources and supported by a digital representation of the analysis structure. This was done to estimate times with unitary inspection data, deterioration mapping control, and data connection to the developed inspection model. A railway tunnel for ore transport in northern Brazil was selected as a case study to apply the forecast model, as it showed an accelerated deterioration process within a few years of service. The tunnel was divided into four sections that were analyzed using the methodology developed through inspection-sensitive analysis. The section with the lowest service level (*TS* 1) matched the geological-geotechnical mapping, which indicated a rock mass with high metamorphic grade and susceptibility to weathering agents. The digital model resulted in an inspection digital twin (*IDT*) prototype that maps the zones with the greatest problems regarding the degradation process. Metadata is assigned to control elements, and there is online connection to a visual database of the tunnel. The target intervention time was set at 16 years by the developed structure, which falls within all forecast intervals calculated by the limit analyses. The inspection *DT* enables digital monitoring of the structure under analysis, allowing for efficient and optimized management.

1. Introduction

Tunnel constructions are known for their long service life compared to other structures, but several factors, such as material quality and the impact of climate change can reduce their durability (Sandrone and Labiouse, 2011; Bakhsi and Nasri, 2019). These infrastructures are specifically designed to facilitate interactions between the structural elements and the geotechnical material, with a service life of up to 100 years, and many transportation structures remain operational for more than half of that period (Hu et al., 2019; Huang et al., 2017; FTA, 1997). Over time, the loads acting on the structure can vary significantly, and the structural changes can result from weathering and environmental factors.

Degradation processes are associated with tunnels, such as chemical interactions between the tunnel and the environment, aging and wastage of materials, besides accidental events such as mechanical shocks, fires, and explosions. These processes are connected to the concepts of robustness and resilience, which refer to the tunnel ability to withstand and recover from damage while avoiding collapses (Stochino et al., 2019).

Structures degradation can disrupt their equilibrium conditions, but effective inspection programs, routine maintenance, and rehabilitation interventions can mitigate their impact. Fig. 1 was developed by adapting representations of level decreases from multiple sources, including Andrade and Martínez (2009), Sandrone and Labiouse (2011), Wang (2018), and Patterson and Perry (1998). These have been updated to reflect an improvement in the maximum quality level. Accurately evaluating performance degradation and forecasting future conditions

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are crucial for implementing maintenance and tunnel lifecycle management strategies (Frangopol, 2011).

The curve depicts the conceptual behavior of the evolution of performance and safety levels over time through structural control processes, which contribute to extending the tunnel service life. Investing in continuous interventions and preventive maintenance can significantly extend the safety limit beyond the originally estimated project lifespan, allowing performance to be restored with each rehabilitation, despite not reaching previous levels. The tunnel can improve its safety levels beyond the initial phase to maximum quality, although achieving this requires more substantial investments than those initially planned during the project phase.

As stated by McKibbins et al. (2009), the gradual decay of tunnel infrastructures can lead to a premature end or reduction in their useful life, resulting in a disruption of traffic flow, economic losses, and the need for costly remediation and replacement works. The structural decay of tunnels can be monitored by verifying the evolution of various responses, such as displacements, leaks, and cracks. These responses serve as indicators of the structural performance degradation.

When evaluating the structural performance of an operational tunnel, it is crucial to accurately determine its service condition and to identify any deviations from the design assumptions (Yuan et al., 2012). Therefore, this paper aims to introduce a methodology for assessing the service state of tunnels and a flexible approach to predicting rehabilitation times. This method is applied to a tunnel located in the northern region of Brazil, which is used for transporting ore via rail. The ultimate goal is to provide insights into the condition of the tunnel and make informed decisions regarding maintenance and rehabilitation interventions to ensure its continued safe and efficient operation.

The digitalization and massive data generation drive the need for updated methodologies in underground infrastructures. Collaborative use of technologies is essential for the current requirements of virtualization, control, and the 'intelligence' of tunnels (Huang et al., 2021). The process of integrating data that mirrors the real-time conditions of a physical system into its virtual counterpart shares certain similarities with the concepts of a digital twin (*DT*) and a cyber-physical system (*CPS*) (Anumba and Roofigari-Esfahan, 2020).

A digital twin is a highly realistic digital replica of a physical product, system, or process. The underlying technology aims to achieve an integrated modeling of the system or product by using asset information to mirror the life of its corresponding twin across multiple scales and uncertainties (Dan et al., 2021; Glaessgen and Stargel, 2012). The resulting digital twin facilitates operation management throughout the asset lifecycle, enabling real-time monitoring and control of its operation (Fig. 2) (Broo et al., 2010).

Recent advances in digitalization have created new opportunities for extracting data, insights, and situational awareness from the performance of physical systems, enabling better management and maintenance of critical infrastructure (Boyes & Watson, 2022). Structural health diagnostic methods based on data are increasingly becoming the norm for structural diagnosis (Housner et al., 1997).

The Operation and Maintenance (O&M) phase is the stage in the life cycle that should ensure proper service, meet user requirements, and maintain reliability throughout the structure's lifespan. The performance, technological control, and diagnostic data of large civil construction assets, such as tunnels, have transitioned from optional to mandatory due to the high costs associated with their malfunction or failure.

The data essential for constructing the virtualization model in this study is acquired through visual inspections, which entail manual processing and integration. Consequently, the absence of automated synchronization justifies categorizing the digital twin developed in this research as an initial prototype. The resulting virtualization adheres to the fundamental principle of digitally connecting the representation to the physical asset, thereby fulfilling the criteria of a DT. Defining the concept remains a significant engineering challenge for the future, as it has not been widely established.

2. Deterioration of tunnels

Deterioration is a common occurrence in all types of constructions, but it poses a particularly significant challenge to infrastructures that must maintain a minimum level of performance over an extended service period as defined in the project. The deterioration process is progressive in nature and is typically associated with the operating time and external conditions. Defects tend to propagate throughout the service life of the infrastructure, as illustrated in Fig. 3.

Fig. 3 illustrates the visual inspection of tunnel defects. The degradation of the structure has reached an advanced stage despite its few years in service, highlighting potential issues in the construction phase.

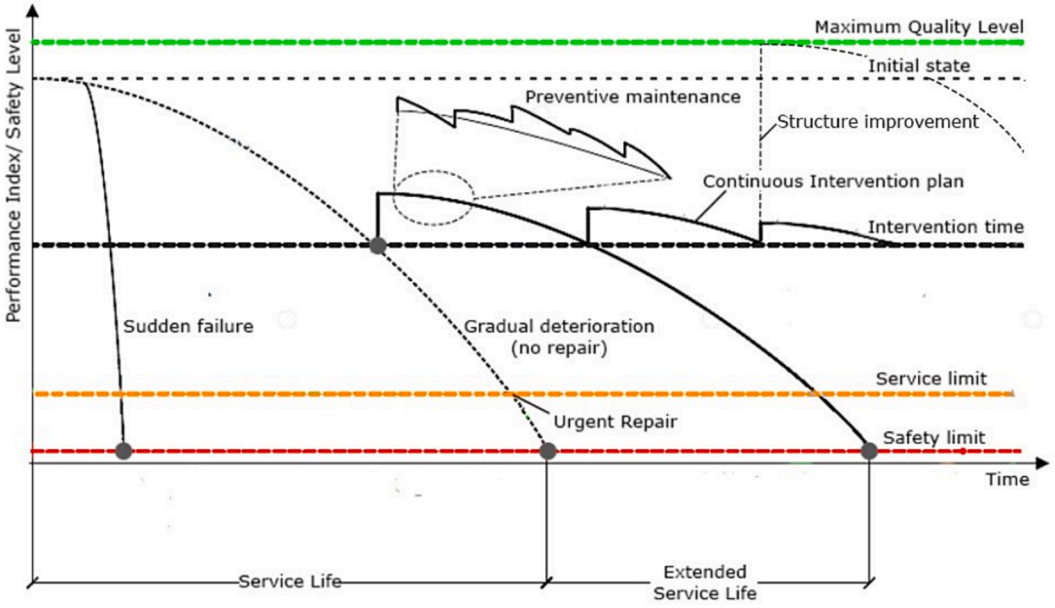

Fig. 1. Time evolution of tunnel quality level and maintenance management ().
Adapted from Sandrone and Labiouse, 2011

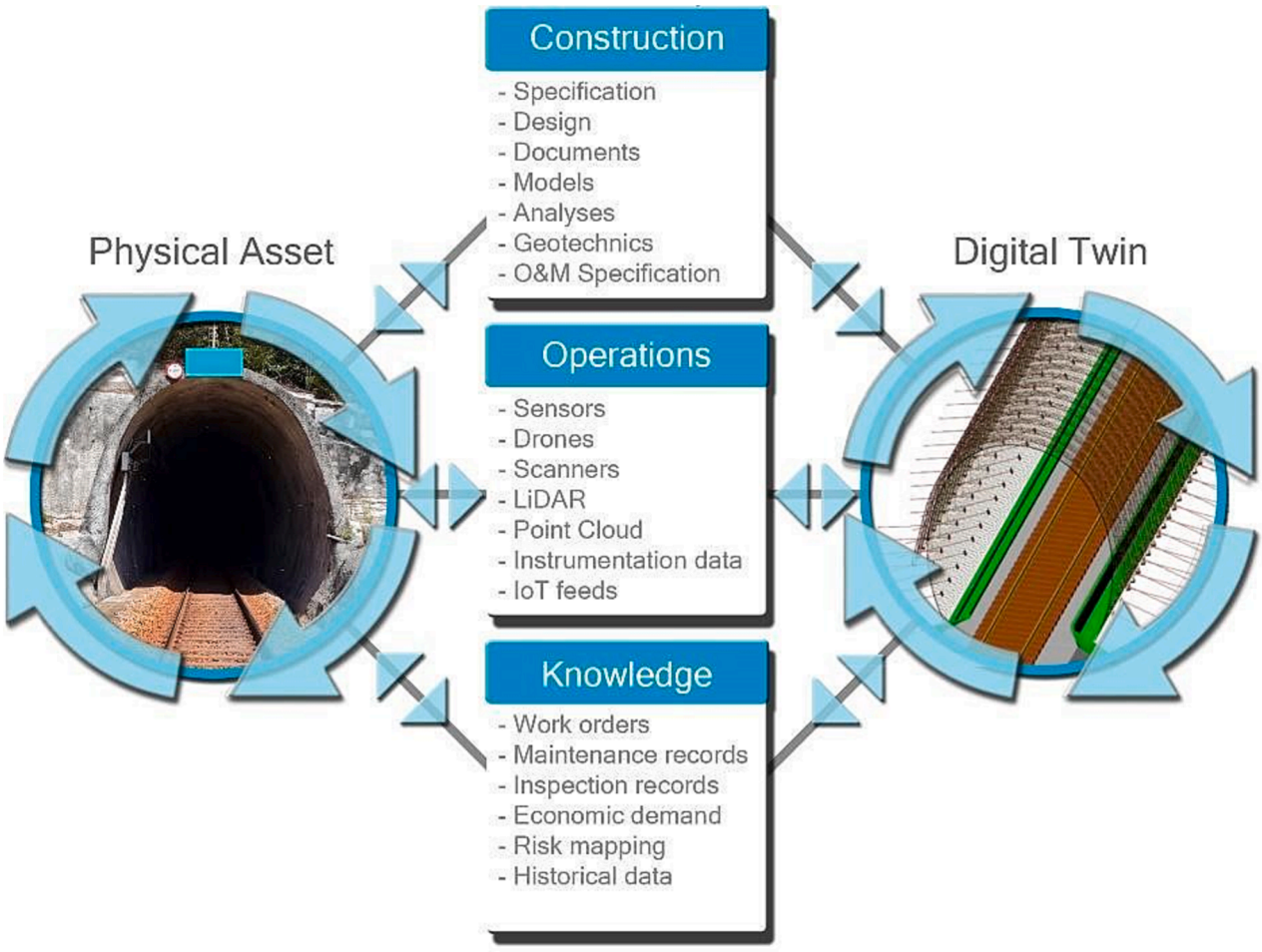

Fig. 2. Conceptual idealization of a digital twin (*DT*) applied to tunnel infrastructures.

Fig. 3. Analysis tunnel defects: (a) cracking and leakage; (b) spalling.

Water-related damage, the primary weathering agent, is the most common type of damage found throughout the length of the lining.

Tunnel infrastructures are often situated in harsh environments and are continuously exposed to water percolation, which is considered one of the major factors in assessing the potential for deterioration of such structures (Prasad et al., 2018). The investigation of performance

degradation processes over time is crucial when designing structures with long service periods and is highly significant for a large number of users.

The classification of defects and service levels into four or five categories is widely used in this type of structure, as noted by Yuan et al. (2012), FHWA (2015), Li et al. (2017), Wang (2018), and Lu et al. (2022). However, there is currently no standardized approach to presenting results and data, each method presenting its own advantages. These include the ability to associate individual deterioration of elements with specific units of measurement, the classification of infrastructure analysis areas into subdivisions, and the quantification of the state of service, among others.

2.1. Defects and service classification

The methodology for generating conditional state data is a fundamental component for predicting the years of service of infrastructure, projecting monitoring campaigns, investing in proper maintenance, and optimizing the life cycle cost of infrastructure (*LCC*). This infrastructure condition is obtained by comparing data from inspection campaigns and monitoring with specific project conditions.

The development of new technologies has increasingly provided greater control of activities in the O&M stage for underground infrastructures (Huang et al., 2021). The use of technologies such as deep learning for recognition and interpretation, as well as point cloud scanning, generates more precise insights into the structure's behavior. Thus, the creation of a DT based on the integration of this new data and historical informations and inspections avaible into models and the generation of a data layer within a structural timeline, as presented in this research, represents a significant advancement for tunnel virtualization.

The development of a maintenance program (Fig. 4) requires quantifying and classifying the degree of degradation and problems in the infrastructure to understand the extent of their impact. Without clear pre-established criteria for distinguishing between different degradation categories, the classification process can become subjective. Therefore, it is crucial to establish clear and objective criteria for classifying infrastructure degradation to ensure consistency in the maintenance program development process.

The control plan for the structure was designed by engineering consultants hired by the tunnel administrators, as featured in the case study of a highway ore transport tunnel in Brazil. However, it was never implemented or the data from these inspection processes were lost. Therefore, the authors aimed to propose an efficient methodology to adapt the results of the available and effectively conducted inspections within the analysis tunnel. The company's database only includes visual defect recognition campaigns and minor maintenance activities performed every 5 years.

The Federal Highway Administration (FHWA) has developed a classification system for evaluating the deterioration of tunnel infrastructure (Table 1). The system assigns each component of the infrastructure to one of four degradation states, based on its condition. The FHWA's specification manual for national tunnels provides guidelines for this classification process. The manual also includes a set of units of measure for each element of the infrastructure, and specifies the fractions of these units that apply to each degradation state. The final report on the infrastructure condition presents this information.

Table 1
FHWA (2015) tunnel defects classification.

Defect	General Condition
State 1	Good condition - no significant distress.
State 2	Fair condition - isolated breakdowns or deterioration.
State 3	Poor condition - widespread deterioration or breakdowns without reducing load capacity.
State 4	Severe condition - The condition warrants a structural review to determine the effect on the strength or serviceability of the element or tunnel OR a structural review has been completed, and the defect impacts the strength and serviceability of the component or tunnel.

Wang (2018) also developed a deterioration classification system, which, as FHWA, comprises four degradation states. However, Wang's system classifies each type of defect (such as corrosion, cracking, delamination, etc.) differently, as outlined in Table 2. In contrast to the FHWA's system, Wang's classification represents the analyzed structure areas, as shown in Fig. 5. The methodology begins by dividing the area under study into a grid composed of smaller regions of equal size.

Table 2
Wang (2018) some tunnel defects classification.

Deterioration	Rebar	Concrete
	Corrosion	Cracking
State I	Fresh or slightly corroded, no intervention is required.	No cracks or cracks are so minor that countermeasures are not needed.
State II	Slight corrosion is superficial or small areas; further monitoring or intervention is required for preventive maintenance.	Fine cracks; because the future function of the structure is likely to decrease, monitoring and measures are required for preventative maintenance.
State III	Corrosion-induced cross-sectional loss in the rebar; slightly reduced structural capacity; measures must be taken as soon as possible.	Cracks are concentrated along the steel reinforcement or some shear cracks; because the structural integrity is reduced, intervention measures should be taken as soon as possible.
State IV	Corrosion-induced cross-sectional loss and significant functional impairment of the rebar; intervention urgently required.	Large-scale crack concentration or shear cracks; because the function of the structure is severely degraded, urgent measures must be taken.

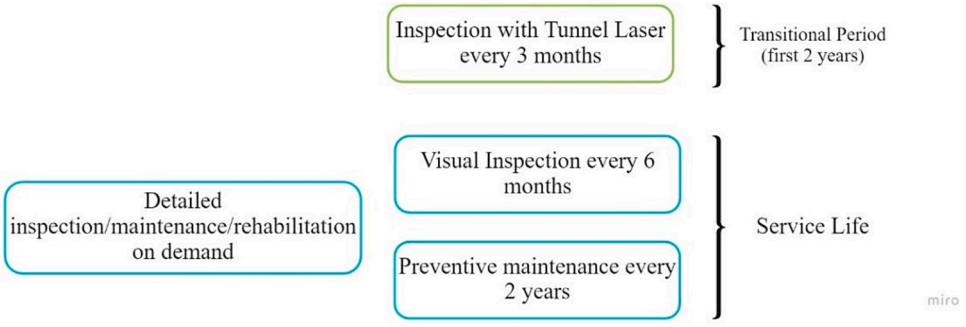

Fig. 4. The inspection and maintenance control plan initially conceived for the case study tunnel.

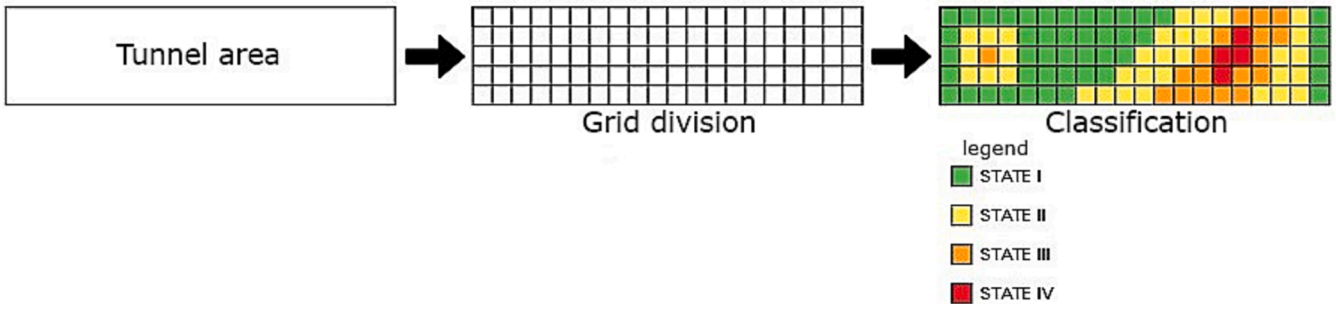

Fig. 5. Deterioration classification in grid methodology.

Different pathologies are then categorized and the area fraction is classified accordingly.

The procedure is significant because it allows evaluating the entire structural system of a tunnel extension for deterioration, as the degradations that trigger instabilities are often not confined to individual elements. This approach yields a mapping of infrastructure risk that is linked to the infrastructure susceptibility to degradation. However, this methodology has limitations in that it does not provide information about the specific processes occurring in the analysis region and the final classification of each grid area. While it uses Table 2 as a reference, it still relies on a certain degree of subjectivity from visual inspection and the inspector's judgment.

To address this limitation, Li et al. (2017) proposed a comprehensive index called the tunnel serviceability index (*TSI*), which quantifies the state of service of a tunnel infrastructure. The author used a partial least squares regression (*PLS regression*) to obtain the *TSI* factor by analyzing monitoring empirical results related to relative and average differential settlements, convergence deformations, infiltration, cracking, and spalling of tunnels excavated in soft soils, as shown in Equation (1). The methodology was applied to the Shanghai subway.

$$\boldsymbol{TSI} = 5.23 - 0.16\sqrt{\boldsymbol{S}_{ave}} - 0.01\boldsymbol{S}_{diff_ave} - 0.09\boldsymbol{C}_{ave} - 0.08\boldsymbol{d}_l - 0.05\boldsymbol{d}_c - 0.50\boldsymbol{d}_s \quad (1)$$

where: S_{ave} is the average settlement between the data collection point with the minor displacement and the other observation points (mm), S_{diff_ave} is the average differential settlement (mm/m), C_{ave} is the moderate convergence strain (‰), d_l is the infiltration area (m^2/120 m), d_c is the crack length (m/120 m), and d_s is the spalling area at a given size (m^2/120 m).

The tunnel infrastructure service efficiency ranges are: *TSI* range 4.0 to 5.0 means a perfect state; from 3.0 to 4.0, the service state is defined as good; 2.0 to 3.0 means a steady state; the range from 1.0 to 2.0 means bad status; and from 0 to 1.0 means terrible condition (Table 3) (Xiao et al., 2020).

The classifications presented by different authors demonstrate varying factors and evaluation methods. Li et al.'s (2017) proposition is based on an empirical and subjective method for calibrating the equation; FHWA (2015) presents a generic classification and the result is a quantitative measure of the unit of analysis; and Wang (2018) promotes an area analysis but loses information on the frequency of specific defects.

This paper aims to address the limitations of existing deterioration and service classification methodologies but incorporating relevant aspects of each. The resulting method fills the gaps and presents a prediction model, described in the next section.

Table 3
TSI limit range (Li et al., 2017).

Status	Rating
Very good	5
Good	4
Fair	3
Poor	2
Very poor	1

2.2. Prediction models

Forecast models are widely used to estimate the time required for maintenance campaigns and infrastructure interventions. However, it is essential to recognize that the systematization and description of future events using calculation models only generate estimated results, and are not exact answers to prediction problems. The complexity of infrastructure interactions and the environment is not fully captured by such models. Therefore, the relevance of results derived from the prediction of tunnel infrastructure aging is directly linked to the quality, quantity of data available, the prediction model to be used and its development.

There are two types of degradation models: probabilistic models and deterministic models. Probabilistic prediction models take into account the randomness of structures and external factors. These models have an advantage in solving problems associated with uncertainties. However, the predicted changes in performance of tunnels using probabilistic models may be subject to randomness (Huang et al., 2017). In contrast, deterministic models can be obtained using mechanics, experiments, or a combination of both. Models that rely on experiments can be influenced by various factors, leading to result variability and inaccurate descriptions of tunnel performance (Xiao et al., 2020).

A tunnel exhibiting a concave performance loss behavior experiences a significant decline in performance within a few years, requiring a shorter rehabilitation time. Conversely, a convex behavior indicates a stabler condition of the structure, with minimal influence from weathering agents and service demands. An S-shaped behavior represents an intermediate situation between the concave and convex behaviors.

Empirical data based on engineering experience and reasonable assumptions suggest that tunnel infrastructure exhibits typical long-term performance loss behaviors (Fig. 6) due to deterioration over time. The method proposed by Xiao et al. (2019) lets us deduce that the loss-in-service behaviors of tunnel infrastructure can be predicted using a model based on an inverse exponential function (Eq. (2).

$$\boldsymbol{TSI} = \boldsymbol{TSI}_0\left[1 - \boldsymbol{e}^{-\left(\frac{t}{\gamma}\right)^{\beta}}\right] \quad (2)$$

where *TSI* is the tunnel service indicator, TSI_0 is the initial service state of the degradation indicator pointing to problems arising from the construction phase, t is the lifetime in service, and α and β are regression coefficients adjusted through monitoring campaigns and inspection data, responsible for the shape of the degradation prediction curve indicated in Fig. 6.

An elevation in the β value leads to a change of the curve, from a concave shape to an S-shaped curve. Similarly, an increase in the α value induces a progression of the curve shape from concave to S-shaped and, subsequently, to convex. Furthermore, Equation (2) provides an opportunity to reconstruct behavior in situations in which historical information is insufficient, resembling an approximate retrieval of

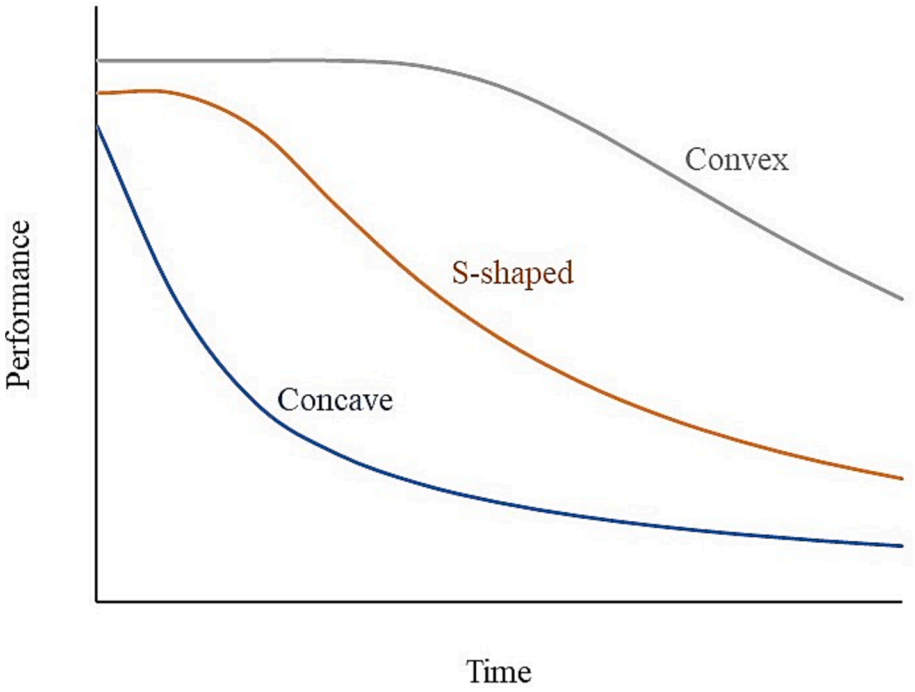

Fig. 6. Tunnel degradation behavior curves. (Xiao et al., 2019).

historical data (Xiao et al., 2019).

3. Proposed methodology

Frequently, inspection and monitoring data in tunnels do not abide by the prescribed design standards. This information may be lost or disregarded due to various issues with the predetermined construction control guidelines. The case study proposes campaigns as shown in Fig. 4; however, only visual inspection campaigns are conducted every five years for infrastructure associated with basic repair and maintenance.

The absence of data hinders the outcomes of prediction models. Forecasting methodologies need to adapt efficiently to the type of data, optimizing the life cycle and control projects of infrastructure based on the available information. Consequently, the subsequent forecasting methodology is illustrated in Fig. 7.

Limitations in *TSI* equation are associated with the absence of crack width as an input parameter for the serviceability index calculation, a factor linked to the initiation and rate of reinforcement and/or fiber corrosion. The shape of cracks has significant relevance in structural behavior (Branco et al., 2010) and degradation models (Zhang et al., 2022), consequently affecting the structural serviceability state. Another important aspect involves errors related to strain measurements, where precision to the millimeters in longitudinal strain input data and per mille (‰) of the external diameter of tunnel in transversal strain input data in Eq. (1) is required. Three out of six variables in the equation proposed by Li et al. (2017) depend on deformation data (S_{ave}, S_{diff_ave}, C_{ave}). In consequence, problems associated with point cloud alignment or convergence measurements could influence significantly the calculation of the *TSI*.

The methodology developed is based on integrating the inspection and classification into four states of equal areas of the structure, as presented by Wang (2018) and FHWA (2015), with the process of state quantification in service developed by Li et al. (2017) and behavior prediction presented by Xiao et al. (2019). This bridge among the processes presented is justified to yield the strengths of each process: defect mapping, state classification, behavior prediction, and structural quantification.

The methodology entails conducting an initial inspection after the first two years of operation. During this inspection, defects are recorded and added to a grid that represents different areas of the tunnel. The grid is then classified based on the quantity or severity of issues within each specific grid cell. This classification process categorizes the degradation of the structure into four states, forming a matrix of classified degradation. Subsequently, an average value of the *TSI* variations is assigned to each degradation cell. By determining an approximate average *TSI*, the outcome of the initial inspection allows calculating TSI_0.

After a three-year period, a follow-up inspection is carried out using the same framework to calculate the estimated average $TSI_{5\ years}$. The data obtained is then used to plot an approximate average *TSI* over a time curve. To conduct a limit analysis, Equation (2) is employed, incorporating in-service state data from the literature, National Tunnel Inventory (NTI) (FHWA, 2022), as well as data from other tunnels within the same railway system that have exhibited similar degradation patterns in their structures (Fig. 8).

The data collected was plotted on a graph of *TSI* in service time, calculated using the approximate methodology through the inspection grid presented later. By employing the inverse exponential equation to model the degradation of service levels throughout the lifespan, lower limit equations were estimated for the different data sources, providing a comprehensive understanding of the overall infrastructure analysis situation.

The four tunnels located along the same railway exhibited similar degradation behavior, as expected, given that they are situated in the same environment and subjected to identical service conditions. The target tunnel fell within the calculated lower limits and was closely aligned with the lower limit curve derived from the data presented in the *FHWA* inventory.

Another aspect to consider pertains to the railway usage of the target tunnel. The target tunnel serves as a mineral transport route, with locomotives passing through it 16 times a day, totaling 5.840 passages per year. Despite the predominantly rock excavation material of Brazilian tunnels and the filtered data from the NTIs, this high-frequency locomotive usage may contribute to the relatively lower *TSI* levels in comparison to highway tunnels.

The literature data, including the *TSI* development research itself,

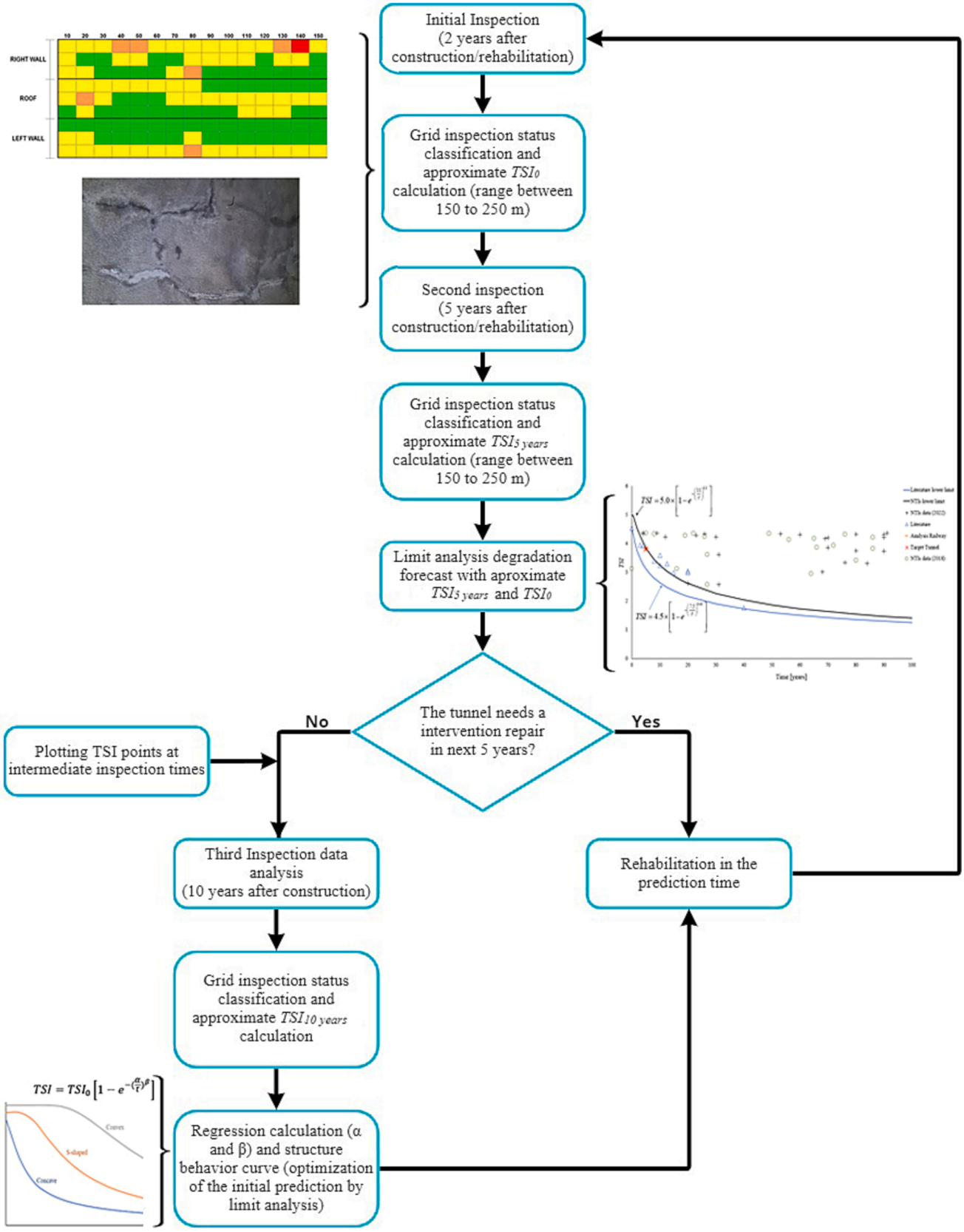

Fig. 7. Forecast methodology framework developed for the analysis tunnel.

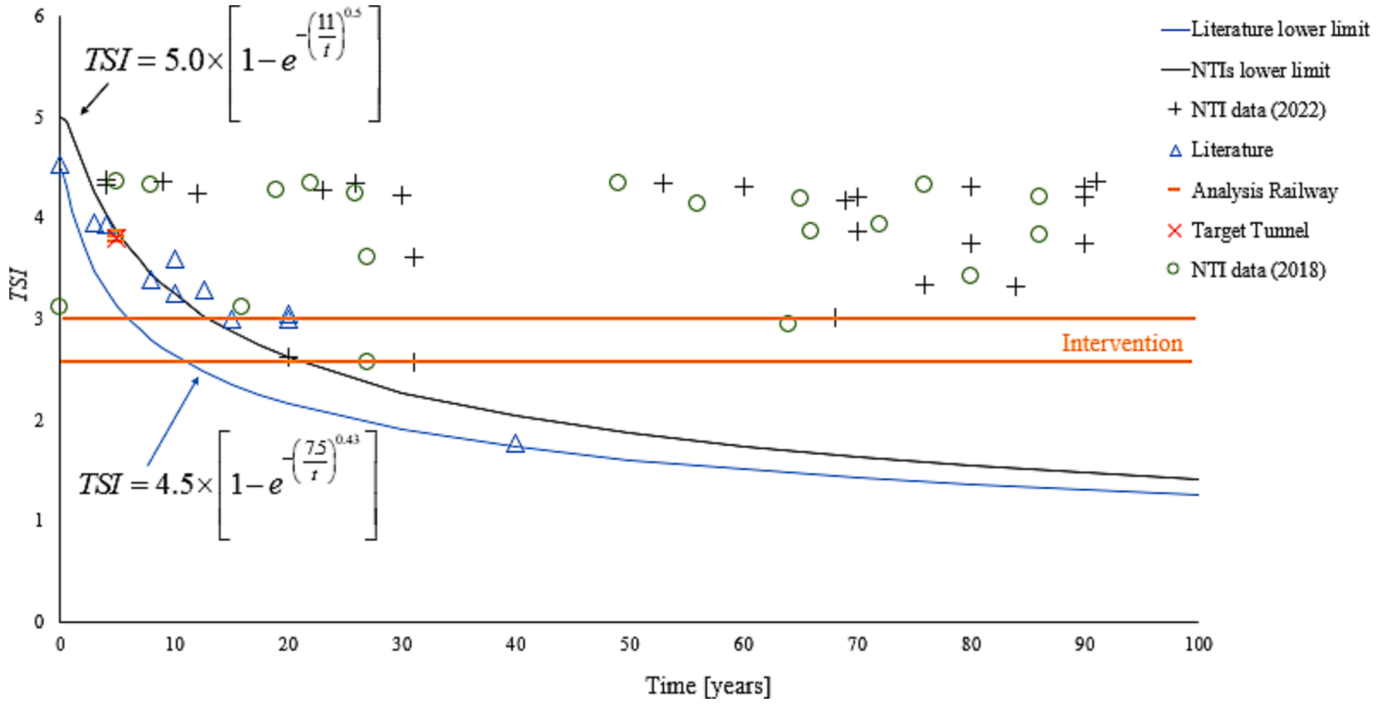

Fig. 8. A limit analysis to degradation forecast for tunnels.

are derived from tunnels excavated in less resistant geotechnical materials, as well as rail tunnels. In contrast, the *NTI* data were filtered to include only tunnels excavated in rock masses, and their usage is exclusively related to road tunnels. The lower limit curves from these two data sources exhibit a more pronounced degradation process in the literature's studied tunnels. Consequently, as indicated by the results, these two factors may influence the *TSI* reduction curves.

The tunnel structures in Brazil are situated in environments that pose challenges to infrastructure due to the humid equatorial climate, characterized by high temperatures and elevated humidity year-round. As a result, the evaluated structures are expected to demonstrate higher levels of degradation in their service states, as depicted in Fig. 8, when compared to other structures outlined in the research data.

The limit analysis aims to understand the degradation behavior of the tunnel based on a single inspection point and compare it to the available surveyed data (Table 4). The analysis area is limited to a range of 150 to 250 m to ensure a satisfactory level of accuracy. These limits are established based on the work of authors that have conducted performance state calculations and degradation predictions in infrastructure systems.

Due to the heterogeneous nature of the researched data, the extraction and organization process was approached specifically for each case. The data from the NTIs (Table 4.b) underwent filtering to ensure a closer match to the target tunnel for applying the methodology.

The filters were based on ground condition (rock), tunnel shape (excluding rectangular), number of bores (1), complexity (non-complex tunnel), lining structure (shotcrete), and tunnel length (≥100 m). The initial dataset consisted of approximately 553 tunnels, and after applying the filters and combining the tunnels that met the criteria, approximately 24 tunnels remained for further analysis.

Certain data in Table 4.b were marked in red due to inconsistencies in the responses, possibly due to degradation. Despite the absence of maintenance records, the tunnel exhibited an improvement in the service level. These divergent results were reevaluated, and the discrepancies are attributed to the subjective nature of visual inspection. This justifies the construction of a digital twin of the structure to enable collective adjustments and enhance classification standardization.

The subsequent inspection campaign in the methodology occurs after ten years of the structure being in service, allowing for calculating $TSI_{10\ years}$. The initial intervention is assumed to be projected to occur after more than ten years of tunnel operation. In cases in which the $TSI_{5\ years}$ values approach the lower limit ($2.5 \leq TSI \leq 3.0$); the forecasts for the intervention period indicate a response with a low limit of less than a decade. Hence, rehabilitation investment is necessary prior to the TSI_0 calculation.

Xiao et al. (2020) suggest intervention when the *TSI* reaches 3.0. However, based on further correlations between degradation classes and service levels, the response in the intervention period is found to better align with maintenance projects. To forecast beyond a ten-year period, the $TSI_{10\ years}$ is utilized to determine the regression and calculate the α and β coefficients, aiming to optimize the prediction and to identify the specific curve shape for the structure.

3.1. Approximate average TSI calculation

The development of a framework for converting the commonly employed classification of tunnel defects into service levels and quantifying the current state of the structure is relevant to the process of prediction and decision-making management. The calculation methodology for an *average TSI* using degradation classification data from inspection campaigns was verified and validated through comparison with stochastic model predictions based on data available in the literature.

The *TSI* was calibrated by employing the *TSR* (Tunnel Serviceability Rating), which represents the average assessment rating of the tunnel structure's condition provided by various professionals (Li et al., 2017). This calibration process inherently involves a degree of subjectivity, akin to the classification method introduced by Wang (2018) in the degradation grid assessment. Consequently, the primary goal is to transform inspection data into quantifiable indicators of structural serviceability and subsequently assess the validity of the inverse exponential prediction equation through a stochastic Markov matrix approach.

Deterioration monitoring data was collected at the 20 and 40-year service milestones of a tunnel infrastructure within a road network, as documented by Wang (2018). Deterministic analyses were conducted to predict infrastructure deterioration using the inverse exponential

Table 4
Tunnel degradation data extraction and compilation: (a) literature and analysis tunnel data; (b) NTIs 2022 and 2018 data.

Tunnel Name or/and Reference	Service start	Analysis lenght (m)	Use	Inspection time (years of service)	Calculated *TSI*
Wang (2018)	1978	150	Immersed tube tunnel	40	1.77
				20	3.05
Shangai Metro Line No. 1 (Chen et al., 2019)	1995	11,200 (Caobao Road to Hanzhong Road)	Metro shield line	12.6	3.30
Xiao et al. (2019)	2010	970	Metro shield tunnel	20	3.00
				10	3.25
				8	3.40
				4	3.94
				0	4.54
Shanghai Metro Line No. 12 (Li et al., 2017)	2013	1400 (Tiantong Road Station to International Passenger Terminal Station)	Metro shield tunnel	3	3.96
Shangai Metro Lines No. 1,2 and 4 (Li et al., 2017)	1990	134,460 (No. 1 + No.2 + No. 4)	Metro shield tunnel	10	3.60
Guangzhou Metro (Wu and Li, 2021)	1997	No informed	Two-way metro tunnel	15	3.00
Analysis Railway (Tunnel ID: 0)	2016	1010	Railway tunnel	5	3.88
Analysis Railway (Tunnel ID: 1)	2016	739	Railway tunnel	5	3.81
Analysis Railway (Tunnel ID: 2)	2016	500	Railway tunnel	5	3.77
Analysis Railway (Tunnel ID: 3)	2016	617	Railway tunnel	5	3.83

(a)

Tunnel Name or/and Reference	Service start	Analysis Lenght (m)	Use	Inspection time (years of service)	Calculated *TSI*
NTI Tunnel ID: 1963	1998 (rehabilitation in 2001)	136	Highway tunnel	20	2.62
				16	3.12
NTI Tunnel ID: 09 0024	1937 (rehabilitation in 2017)	362	Highway tunnel	4	4.38
NTI Tunnel ID: 53C0251	1951	194	Highway tunnel	70	4.21
NTI Tunnel ID: 53C0601	1941	145	Highway tunnel	80	4.32
				76	4.33
NTIs Tunnel ID: 8140003P0000000	1918 (rehabilitation in 1994)	820	Highway tunnel	26	4.35
				22	4.35
NTI Tunnel ID: 8800034P0000000	1930	635	Highway tunnel	90	4.22
				86	4.22
NTI Tunnel ID: 1378001P0000000 and 1378003P0000000	1936	234	Highway tunnel	84	3.32
				80	3.42
NTI Tunnel ID: D-15-AS	1953	107	Highway tunnel	68	3.02
				64	2.95
NTI Tunnel ID: F-15-AW, F-15-AX, F-15-AY, F-15-X and F-15-Y	1945	1134	Highway tunnel	76	3.34
				72	3.94
NTI Tunnel ID: 003TDIAH0900030	1940 (rehabilitation in 2020)	136	Highway tunnel	80	3.75
NTIs Tunnel ID: 009,000,300,302,596	1951	97	Highway tunnel	70	3.86
				66	3.87
Tunnel ID: T01040200000001	1931	128	Highway tunnel	90	4.31
NTI Tunnel ID: 6700001P0000000, 6700002P0000000 and 6700003P0000000	1938 (rehabilitation in 1990)	154	Highway tunnel	30	4.23
				26	4.25
NTI Tunnel ID: 000,000,000,000,001	1949 (rehabilitation in 2017)	161	Highway tunnel	4	4.34
				0	3.13
NTI Tunnel ID: 5140135P0000000	1940 (rehabilitation in 1990)	107	Highway tunnel	31	2.58
				27	2.58
NTI Tunnel ID: 5140136P0000000	1938 (rehabilitation in 1990)	107	Highway tunnel	31	3.62
				27	3.62
NTI Tunnel ID: 02,552	1940	235	Highway tunnel	9	4.37
				5	4.37
NTI Tunnel ID: 03437)	1932	332	Highway tunnel	23	4.28
				19	4.29
NTI Tunnel ID: 03,961	1931	218	Highway tunnel	90	3.75
				86	3.84
NTI Tunnel ID: 1590012P0000000	1930	1711	Highway tunnel	91	4.37
NTI Tunnel ID: 0002093A0000000	1936 (rehabilitation in 1968)	241	Highway tunnel	53	4.34
				49	4.36
NTI Tunnel ID: T-BKL	1908 (rehabilitation in 1952)	101	Highway tunnel	69	4.18
				65	4.19
NTI Tunnel ID: T-BZQ and T-BZS)	1960	1057	Highway tunnel	60	4.31
				56	4.14
NTI Tunnel ID: 2247)	1930 (rehabilitation in 1998)	686	Highway tunnel	12	4.24
				8	4.34

(b)

equation proposed by Xiao et al. (2019).

To convert the deterioration classification values provided by Wang (2018) into the tunnel serviceability index (*TSI*) proposed by Li et al. (2017), criteria were established. The conversion process allowed for calculating the *TSI* based on the observed deterioration, enabling future projections of the tunnel service index and estimating long-term degradation. These calculations provide insights into the degradation trends over time.

The monitoring data for the tunnel structure was quantified and linked to the corresponding deterioration classes identified using the categorization process. The entire monitoring area of the tunnel floor slab, measuring 4.5 m x 150 m, was divided into 225 elements that experienced significant deterioration over the 20-year service cycles.

Using the average *TSI* values associated with each degradation class from Table 4, along with the limits established by Li et al. (2017) from Table 3, and considering the number of elements in each class for different years, factors were calculated for the respective years of the infrastructure inspection data collection campaigns (as shown in Table 6). This approach allows quantifying the tunnel performance in different service periods by adjusting the curve using an inverse exponential equation.

The calculation of the tunnel service factor involved quantifying elements in different states of degradation and assigning tunnel service factor values to the corresponding deterioration classes. This analysis revealed a significant decrease in the performance of the infrastructure over a 40-year period. Note that the maximum service limit was assumed for the analysis at the start of operations ($TSI_0 = 5.0$).

Applying Equation (2), the α and β coefficients that provide the best fit to the calculated *TSI* values were determined through regression analysis (refer to Table 5). The coefficients obtained for fitting the service factors are 19 for α and 1.1 for β, indicating an S-shaped curve. The curve was divided into states intervals based on the limits proposed by Li et al. (2017) (Fig. 9.a) (0 to 5).

The stochastic curve presented in Fig. 9.b was developed by Wang (2018) using a Markov matrix process, which employed a probabilistic state change matrix to model the structure service behavior in time. The figure depicts the number of elements in each state throughout the lifespan of the structure, providing valuable insights into its degradation progression.

When comparing the curve obtained from deterministic calculations with the stochastic predictions based on the deterioration matrix presented by Wang (2018) (Fig. 9.b), several observations can be made. The transition from State I to State II closely aligns with the upper limit forecasted by the Markov curve for State I. The maximum deterioration in State III coincides with the point where the TSI degradation curve transitions from State II to State III. Furthermore, the endpoint of maximum growth in the State IV deterioration curve corresponds to the period when the curve transitions from Stage III to Stage IV were obtained.

The results demonstrate a robust correlation between the stochastic and deterministic calculation methods, emphasizing the reliability of using the *TSI* as a key indicator for assessing the condition of the tunnel. A comparative calculation was conducted to streamline the conversion process of the categorized elements in the deterioration state grid into estimated mean *TSI* values.

An important observation regarding the conversion methodology developed, tested for compatibility with alternative prediction approaches, lies in the change of maximum limits. According to the methodology, a tunnel that has all its elements in State I or State IV would be classified with *TSI* values of 4.375 and 0.625, respectively, with intervention estimations based on an interval rather than a fixed *TSI* value (Fig. 9). This approach allows for structural improvement within the upper limit, as shown in Fig. 1, while the lower limit may continue with the degradation process beyond the methodology boundary.

Table 5
Assignment of *average TSI* intervals to the states of deterioration.

Deterioration	*average TSI*
State I	4.375
State II	3.125
State III	1.875
State IV	0.625

Table 6
Degradation category elements and calculated average *TSI*.

Time	State I	State II	State III	State IV	*TSI*
Initial	225	0	0	0	5.000
20 years	74	63	88	0	3.047
40 years	25	21	89	90	1.769

4. Case study and inspection digital twin model

The presented methodology was applied using data from visual inspections conducted by the railway company in the northern region of Brazil. The railway connects the northern and northeastern regions of the country for ore transportation (Fig. 10), encompassing various rail infrastructures. Among these, Tunnel 01 was selected for the application of the framework and the creation of the digital inspection twin.

4.1. Analysis of a railway tunnel

The target tunnel, renamed to Tunnel Laboratory 01 (*TL 01*), exhibits significant damage to its shotcrete lining structure, despite its few years of operation (Fig. 11). With a length of approximately 740 m on a single track, even minor disruptions result in the complete cessation of operations. The selection of this tunnel was based on a comparison of the degradation levels among different tunnels, its geographical location, and technical site visits to assess its condition.

The infrastructure is constructed using the drill and blast method, with a lining composed of shotcrete reinforced with fibers. The significant damages observed are linked to water infiltration, corrosion and oxidation of the mesh or metallic reinforcement, as well as cracking in the lining. Certain areas of the inspection grid exhibit two instances of deterioration, while one particular area, there are three instances associated with observations and inspection reports as documented, leading to its classification as state IV.

The geological mapping is depicted in Fig. 12. The image illustrates the various layers comprising the local geology, with significant proportions of metamorphic rocks, as well as regions displaying heightened vulnerability to weathering agents in proximity to the northern entrance. The mapping highlights structural alignments of the rock formations, groundwater levels, and the tunnel excavation.

As a result, excavations in metasandstone and soil are expected to present higher levels of weathering. The *TL 01* is thus divided into three sections (*TS*) of 200 m and one of 140 m for applying the methodology.

4.2. Inspection digital twin (IDT)

The creation of a digital twin for inspection purposes is linked to the advantages of three-dimensional visualization of the structure, assignment of inspection metadata, efficient processing, management of results, and connection with online databases for real-time information directly from the field. These aspects serve as a bridge connecting data from the physical infrastructure to the digital model, enabling the possibility of a virtual tunnel inspection.

The *IDT* construction helps understanding the behavior of the infrastructure, as proposed by Andrade and Martínez (2009). It also orchestrates and optimizes the decision-making process, such as the

Table 7
Results of methodology calculation.

TS	Elements	State I	State II	State III	State IV	$TSI_{5\ years}$	Intervention start [years]
TS 1	180	93	79	7	1	3.708	9.5 to 11.5
TS 2	180	110	60	10	0	3.819	11.5 to 13
TS 3	180	112	64	4	0	3.875	11.5 to 13
TS 4	126	73	52	1	0	3.839	11.5 to 13
Total	666	388	255	22	1	3.810	–

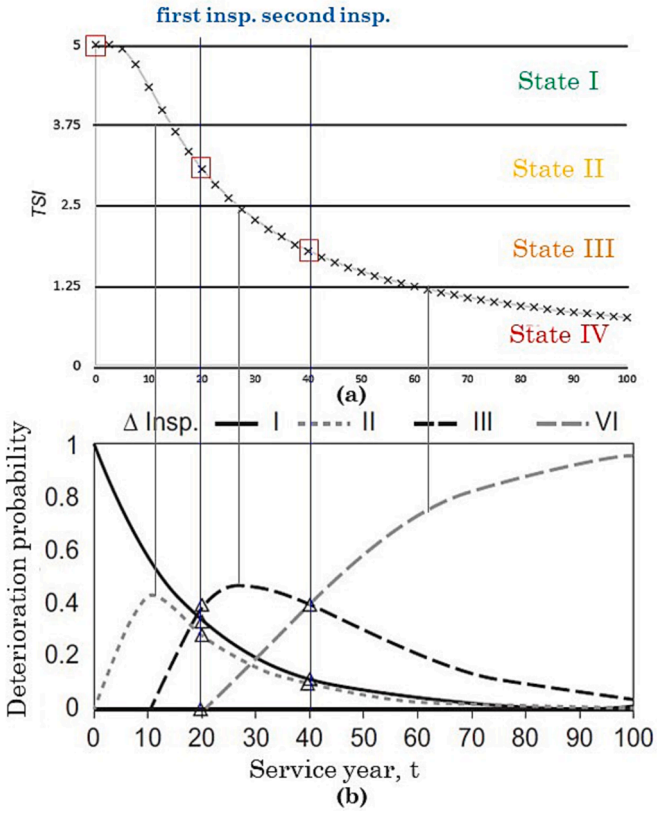

Fig. 9. Predictions compatibility: (a) Methodology curve analysis; (b) Stochastic matrix analysis (Wang, 2018).

need for rehabilitation, as presented in the proposed methodology (Fig. 7). Virtualizing the tunnel leverages the data derived from the sensitive inspection process to align with modernity and Industry 4.0 principles, thereby reducing the loss of information obtained and updating the life cycle management.

The digitization of as-built data for the infrastructure was carried out using the computational tool Rhinoceros 3D (Fig. 13), a proprietary three-dimensional modeling software based on the Non-Uniform Rational Basis Spline (NURBS) technology. This approach aims to minimize approximations for short-term processing while ensuring a high level of detail and metadata attribution required for the representation in a digital twin.

The EleFront plugin for Grasshopper, the visual programming editor of Rhino3D, enables implementing metadata in structural elements and the inspection grid. This metadata, represented as attributes, helps organize and manage large models with numerous objects. It enables users to query, sort, and filter objects within the Grasshopper environment and workflows.

In the virtualization of *IDT* elements, metadata is assigned in key-value format, enabling connection with online databases. Online inspection photo connections were assigned to digital twin elements using Google Photos, allowing the storage of sensitive inspection information.

Fig 10. The location of the railway where the analysis tunnel is integrated.

The selection of the online storage tool was primarily based on preexisting information available on Google Drive, facilitating data transfer.

To depict the inspection campaign and the proposed methodology as shown in Fig. 7, the model was adjusted to preserve the structural elements and topography while incorporating an inspection and classification grid (Fig. 14). Each grid cell would have dimensions of approximately 2 m in width (The total arc is divided into 9 parts, with 3 sections for each: the right wall, the left wall, and the roof.) and 10 m in length. This specific size aligns with the divisions already employed by the management company in sensory inspections, facilitating the company's adaptation to this methodology.

The objective is to enable the inclusion of metadata regarding the elements undergoing field testing and to establish connections between the conducted inspection campaigns. The data from the grid plays a crucial role in directing the execution of tests, with their outcomes being integrated into the model.

5. Results and discussion

The results of visual inspection campaigns were used to classify the four study sections, as illustrated in Fig. 15.a. The classification involved assigning State I to areas without defects, State II to sites with one fault perpendicular to at least two State II areas or one State III or IV area, State III to areas with two pathologies, and State IV to areas with three defects.

To count the number of defects in each category, a Visual Basic for Applications implementation was utilized to tally the cells in the grid corresponding to each degradation state. A total of 666 cells were analyzed across the four grids, with slightly over half of the areas

Fig. 11. Target Railway Tunnel: (a) Analysis Tunnel Laboratory 01 (*TL* 01); (b) Defects in tunnel walls.

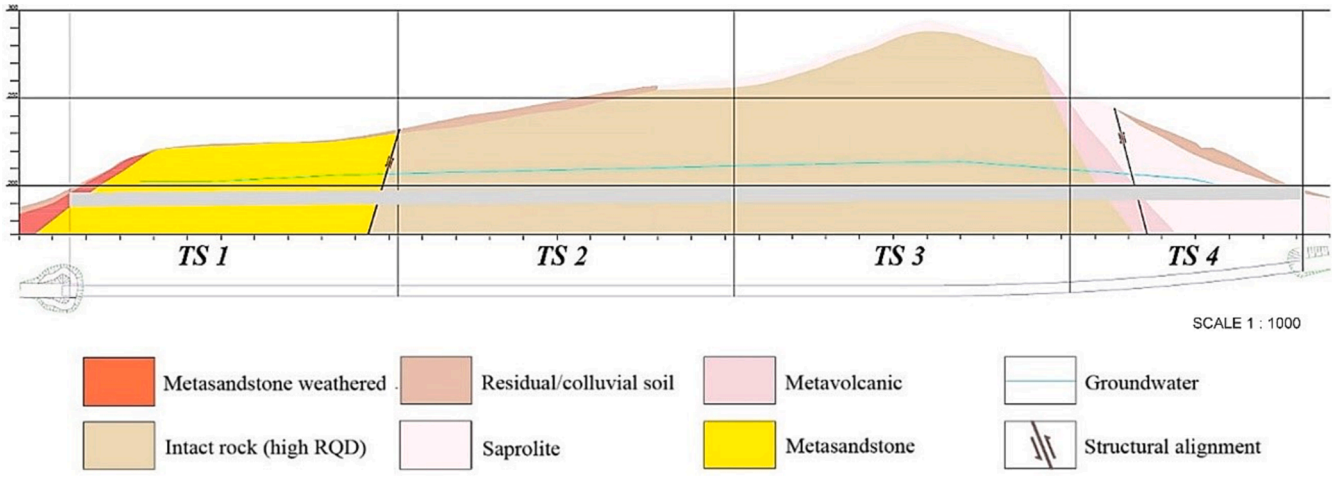

Fig. 12. Geological and alignment data with sections division.

determined to be in good condition. The inspections were conducted on the right and left walls, as well as on the roof of the tunnel.

Fig. 7 illustrates this step as the verification of status and classification for calculating the TSI_{5years}. Due to the lack of information regarding an inspection focused on detecting construction failures and determining the TSI_0 (initial *TSI*), this particular step is unviable in the case study.

5.1. Sections analysis

The first section, located closer to the north exit, is excavated in sandstone material. This material exhibits a high degree of weathering, particularly near the tunnel entrance, as depicted in Fig. 16. The geological mapping confirms that the rock in this area is highly susceptible to weathering agents. Consequently, when applying the methodology, the results are expected to indicate lower service levels for this section, which indeed occurs. However, with the right wall having an average *TSI* of 3.542 (Table 8), the factor is still not low enough to strongly suggest the need for an intervention range. It may qualify for investments in preventive maintenance.

The classification applied to the structure inspection grid reveals a unique state IV classification area where there is a simultaneous presence of reinforcement exposure, cracking, and infiltration on the right wall of the tunnel. The classification indicates that most grid elements in state II are associated with defects in close proximity to or adjacent regions with more severe structural damage.

The *TS* 2 and *TS* 3 sections are located in a rock mass that is structurally competent and has a high Rock Quality Designation (*RQD*). These sections are anticipated to demonstrate superior performance, primarily because they are composed of a robust and resilient solid material. This solid material reduces the influence of flaws caused by loading conditions and minimizes the detrimental effects of weathering agents and deformations.

The *TS* 2 section is fully embedded within a highly competent material, which is expected to result in a substantial increase in the *TSI* value when employing the methodology (Table 7). The deterioration

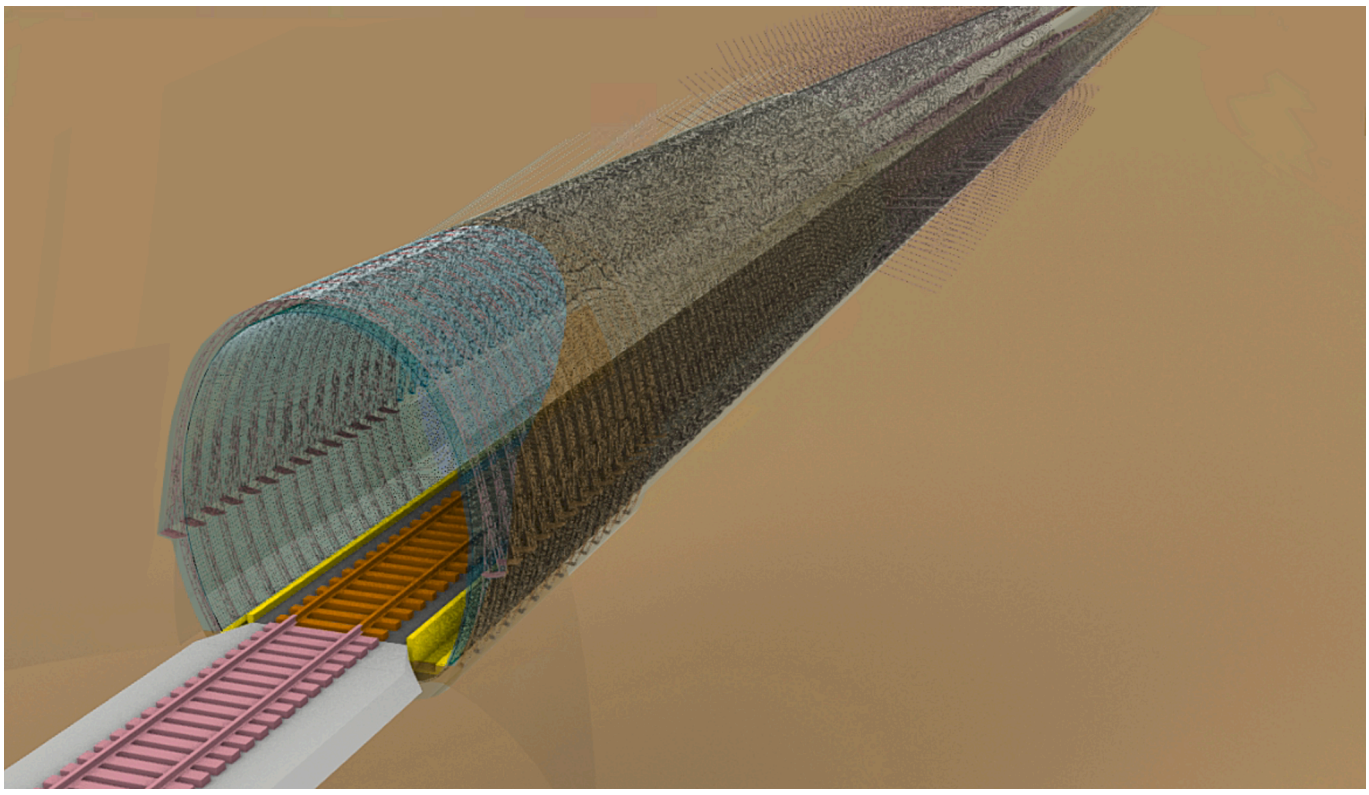

Fig. 13. Digital representation of the *TL 01* in Rhinoceros model.

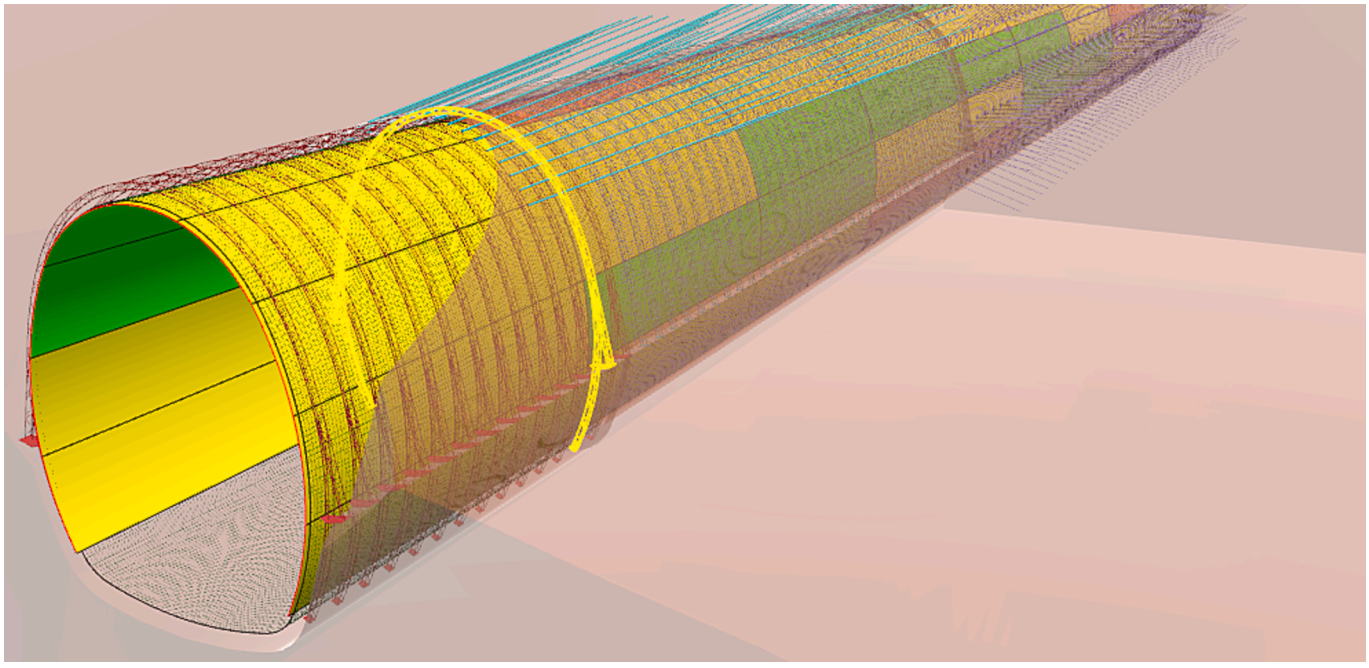

Fig. 14. Inspection Digital Twin (*IDT*) for *TL 01*, inspection grid and structural elements (lattice girders, reinforcing mesh and rock bolts).

matrix classification indicates a greater number of elements in level III, but shows significant improvement when compared to the *TS* 1 grid.

The *TS* 3 section is anticipated to exhibit the highest performance levels among the entire structure. This is attributed to several factors, including its non-adjacency to the *meta*-sandstone, increased confinement resulting from greater tunnel coverage, and the presence of a competent rock mass surrounding it. The methodology demonstrates significant reductions in level III when compared to *TS* 2, as well as the highest number of states classified as level I in comparison to all the other sections, as show in Table 7.

Comparing rating matrices 2 and 3, a significant improvement in service levels can be observed for the latter section. This improvement can be attributed to its significant distance from materials with high levels of alterability, as well as the greater confinement provided by the

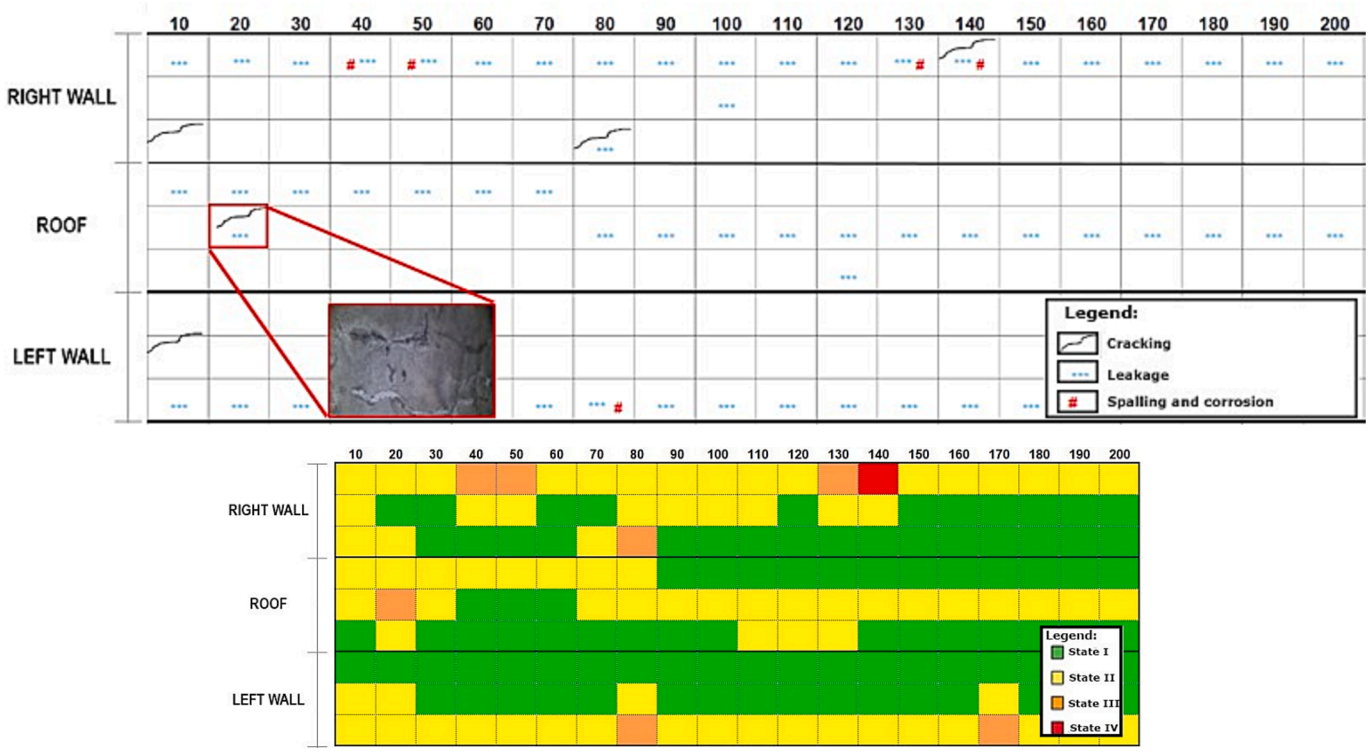

Fig. 15. Data from the *TL 01*: (a) tunnel defect grid; (b) grid deterioration classification.

increased cover.

The final section has the shortest length of analysis, spanning 140 m with 126 elements. The geological composition includes saprolite material, iron deposits, and altered volcanic rock. A road intersects the tunnel perpendicularly in the superficial soil layer.

Due to the greater exposure to weathering at the ends of the excavation, a reduction in performance is expected, as compared to the previous sections. Additionally, *TS* 4 is embedded in a less competent material within the rock mass when compared to the preceding section.

The classification methodology agrees with the predicted deterioration behavior of the structure when comparing the defect grids derived from visual inspections. The right wall is observed to frequently exhibit more significant issues compared to the other analyzed.

5.2. Average TSI and forecast time intervention

The methodology described in section 3.1 was utilized to convert the classification grid of deterioration states into average performance indices (TSI_{5years}), as depicted in Fig. 17. The application of this methodology provides crucial information for asset management and facilitates the optimization of decision-making plans.

The calculated values do not indicate critical states requiring immediate intervention ($TSI = 3.0$). However, the *TS* 1 region can be concluded to be the most vulnerable to degradation and service loss within the structure, while *TS* 3 exhibits better performance. These calculated values can be effectively utilized to manage maintenance investments efficiently.

Consequently, the entire structure is in an acceptable service condition ($TSI > 3.0$), requiring minor repair and predictive maintenance, but the railway infrastructure does not require immediate intervention (Fig. 1). Therefore, the railway company has opted for a campaign of localized and straightforward repairs and maintenance given the current state of the structure, which has been in service for approximately 5 years.

Data from the initial post-construction inspection, as outlined in the tunnel maintenance plan and shown in Fig. 4, were not available. This absence of data poses a significant challenge to the application of the methodology for calculating TSI_0. The lack of data is an important consideration within the flexible nature of the intervention forecasting framework developed.

The prediction process guided by the collected deterioration data involved determining steps for the different parameters that influence the curve (α, β, and TSI_0), limited by the variation of the lower limits identified in the data. This approach resulted in the creation of auxiliary curves, known as curve fitting models, which spanned from the literature limits to the NTIs. The average *TSI* values calculated for different sections of the tunnel were plotted (Fig. 18).

The indices plotted will be assessed using fitting curves to determine the projected timeframe for structural rehabilitation. The *TS* 1, representing the lowest service index value among the sections, closely aligns with the fitting curve that closely matches the NTIs lower limit. This specific fitting curve has curve shape parameters α, β, and TSI_0 of 10.3, 0.486, and 4.9, respectively. While the index falls within the curve, a conservative approach is taken to determine the range of projected rehabilitation start periods between the limiting fitting curves: the lower limit and the coincident curve. A similar methodology is employed for *TS* 3, which aligns with the lower limit of the NTIs (Table 7).

The responses of concave-shaped forecast curves resulted in significant time intervals between the projected start ($TSI = 3.0$) and end ($TSI = 2.5$) of infrastructure intervention, with an average variation of 8 years between the limits analyzed, due to a single inspection data. Despite the distance between these limits, this period can be used as a justification for increased investment in monitoring and condition assessment of the tunnel during this phase of its lifecycle.

Table 7 summarizes the analysis and suggests the potential for initiating interventions after approximately 12 years, with the intervention intervals encompassing the anticipated service life of the tunnel. This estimation allows identifying a target intervention time (Fig. 19). However, it is essential to maintain regular inspections and to continually calibrate the data collected to establish a specific trendline for this particular tunnel. Additionally, intermediate repairs focused on the initial section of the north entrance may be necessary around 15 years of

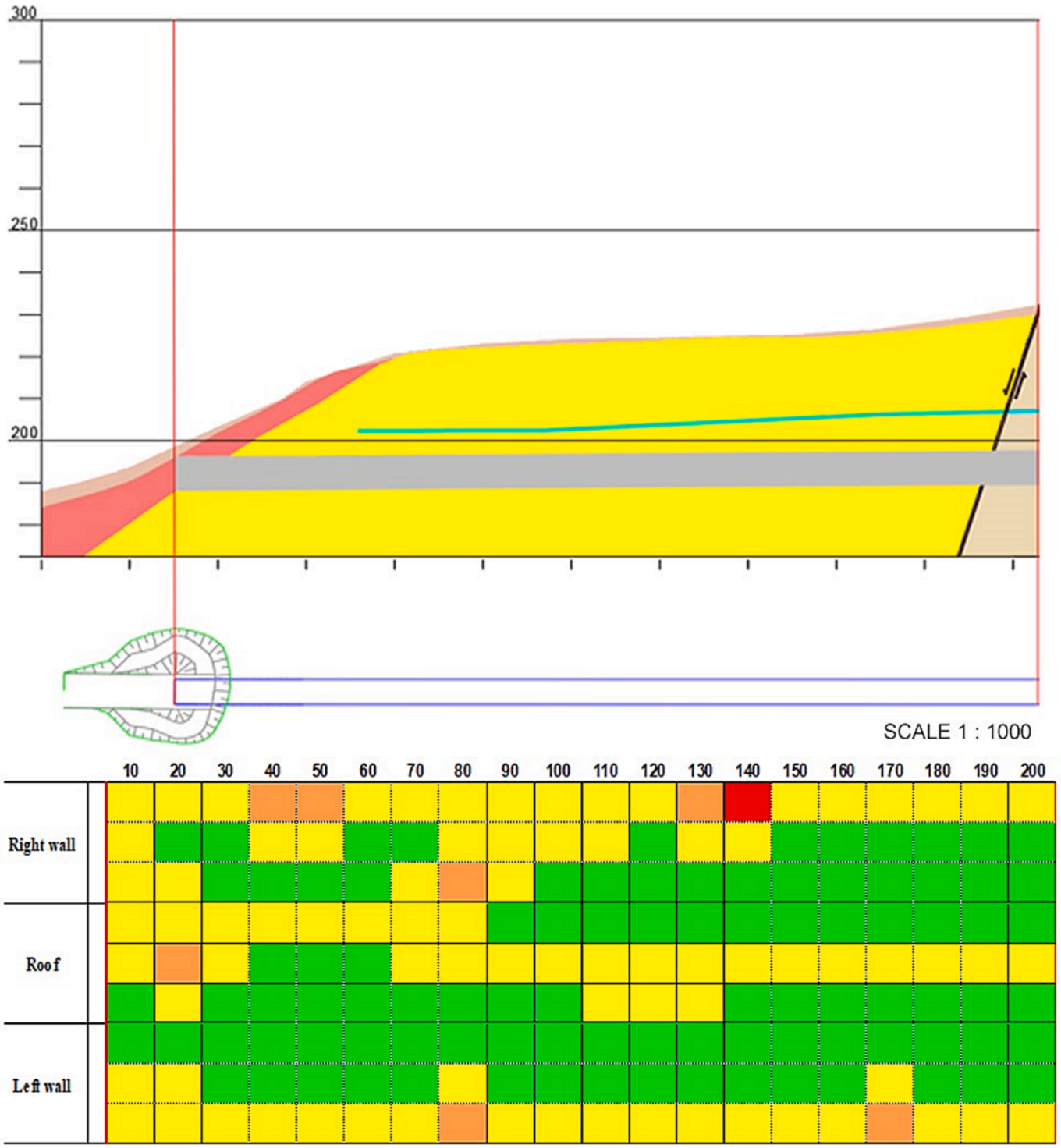

Fig. 16. Grid defects classification and alignment for *TS* 1.

Table 8
Average $TSI_{5\ years}$ for different tunnel areas.

Area	*TS* 1	*TS* 2	*TS* 3	*TS* 4
Right wall	3.542	3.667	3.875	3.810
Roof	3.750	3.958	3.792	3.929
Left wall	3.833	3.833	3.958	3.780

service for the structure.

The analysis section examined various areas using the methodology developed. Among all the sections analyzed, the right wall exhibited the lowest performance index values (Table 8). This finding highlights the need for focused attention and targeted efforts in this specific area. By identifying regions with significant problems, the local analysis enables the optimization of monitoring plans and maintenance processes. This approach ensures that resources are allocated efficiently, allowing for timely interventions and improved performance in critical areas of the tunnel.

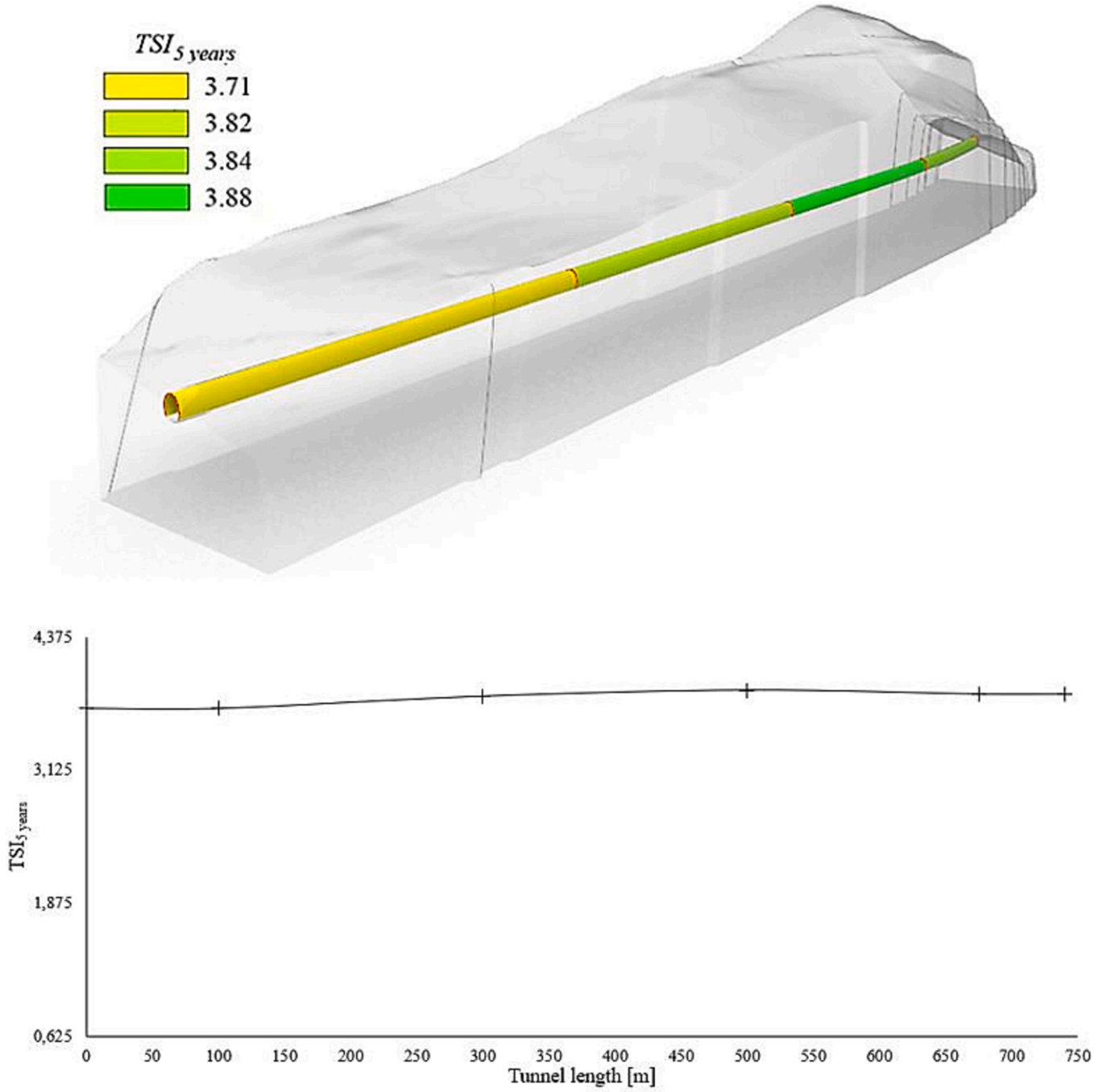

Fig. 17. $TSI_{5\ years}$ analysis tunnel classification and tendency service index in structure length.

In the *TS* 3 section, the service levels of the walls and ceiling are closely aligned. This can be attributed to similar factors influencing their condition, as discussed previously. However, in the *TS* 1 region, there is a noticeable disparity between the index of the right wall and that of the roof and left wall. This indicates that the *TS* 1 region is more susceptible to the deterioration process of the structure. It highlights the importance of a rational selection of sensors and their proper placement in the infrastructure, as emphasized by Tondini et al. (2015). Strategically locating sensors makes it possible to effectively monitor the condition of the *TS* 1 region and to identify any deterioration or potential issues at an early stage, enabling proactive maintenance and preservation efforts.

Due to the recent construction of the tunnel and its estimated project lifetime of 100 years, the methodology could not be fully tested in subsequent stages. However, preventive maintenance activities were carried out this year as the *TSI* did not indicate the need for intervention in the structure, according to the framework presented in Fig. 7.

5.3. Inspection digital twin (IDT) performance

Tunnels are complex infrastructures that generate spatiotemporal data from various sources (Wu et al., 2022). This data is often fragmented and heterogeneous, consisting of project documentation, as-built information, photos, text documents, spreadsheets, diagrams, and design files. Effectively managing, processing, and interacting with this data is crucial for building a robust *DT* of a tunnel.

The development of the *IDT* has brought a significant enhancement to the infrastructure life cycle management. It has allowed for cataloging specific damages, guiding targeted repair campaigns, standardizing

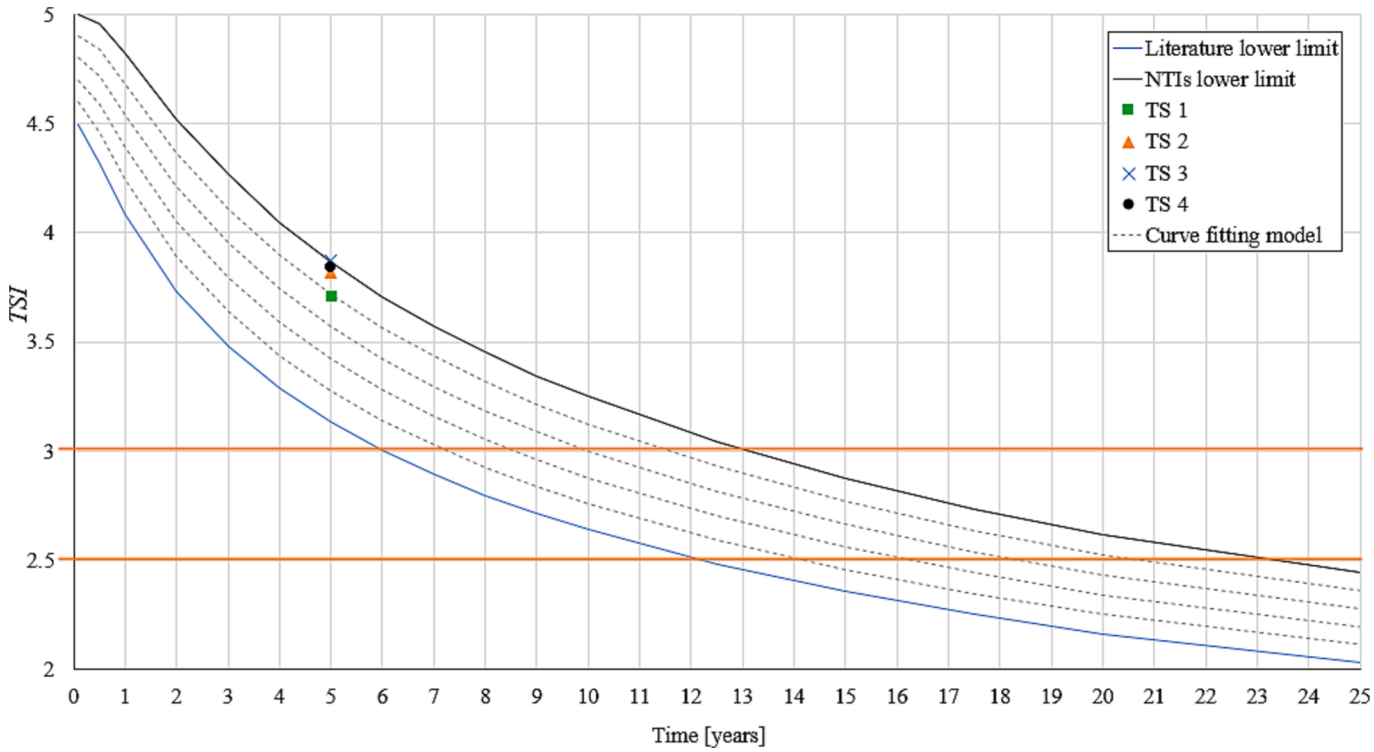

Fig. 18. Limit analysis application for all *TL 01* sections and fitting curves.

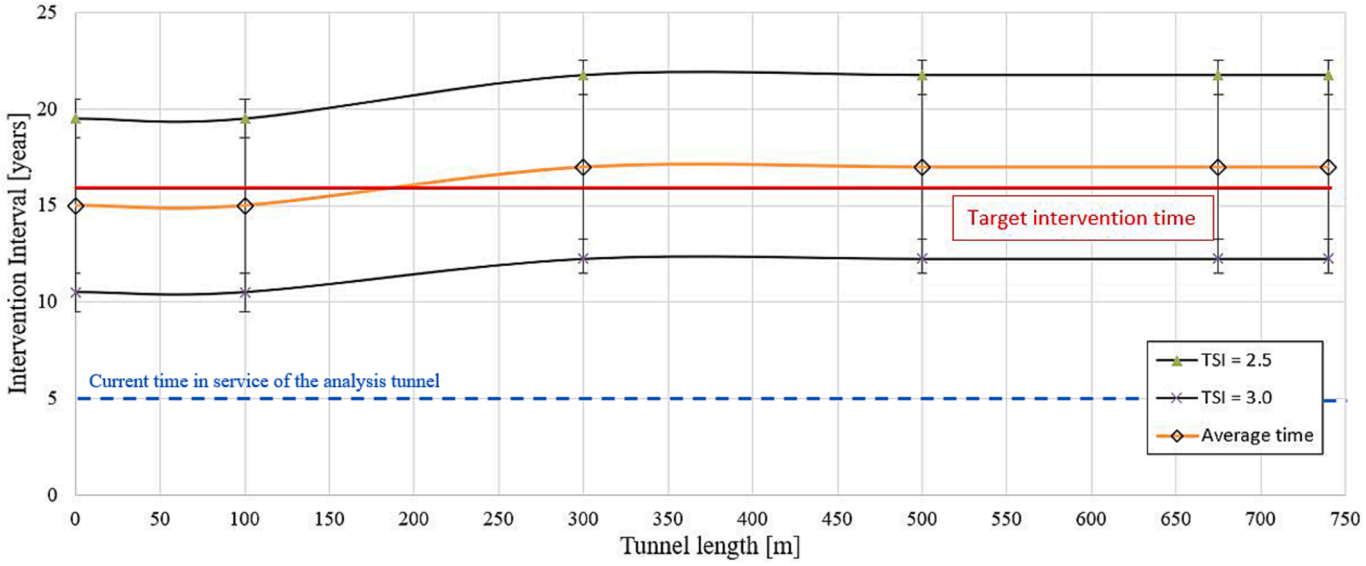

Fig. 19. Target intervention time in the *TL 01* extension for developed prediction model.

knowledge on structural defects, and creating a database of information generated by the methodology developed (Fig. 20). The digital twin provides a comprehensive and structured platform for monitoring and analyzing an infrastructure condition, facilitating informed decision-making and optimizing maintenance strategies.

The Rhinoceros model was created using the as-built data from *TL 01*, primarily concentrating on modeling structural elements and a 3D grid with sensitive inspection information to form the *IDT*. Elements were categorized into layers (Table 7), and critical areas and elements were linked to metadata to organize various data types through the EleFront plugin. This included project documentation, images, and design files, enabling easy access throughout the tunnel's life cycle stages (Fig. 20). This approach facilitated the creation of a virtual representation of the tunnel connected to its inspection data.

Cloud data storage ensures accessibility, security, and scalability. These data were linked to the model through hyperlinks to web addresses. The real-time reports, generated from processed data, provide essential predictive maintenance and repair recommendations to support the established framework. Collectively, these components empower informed decision-making and efficient tunnel infrastructure management.

The management strategy involves collecting, organizing, and integrating different data sources into a centralized database or information management system. This allows for easy access and retrieval of relevant

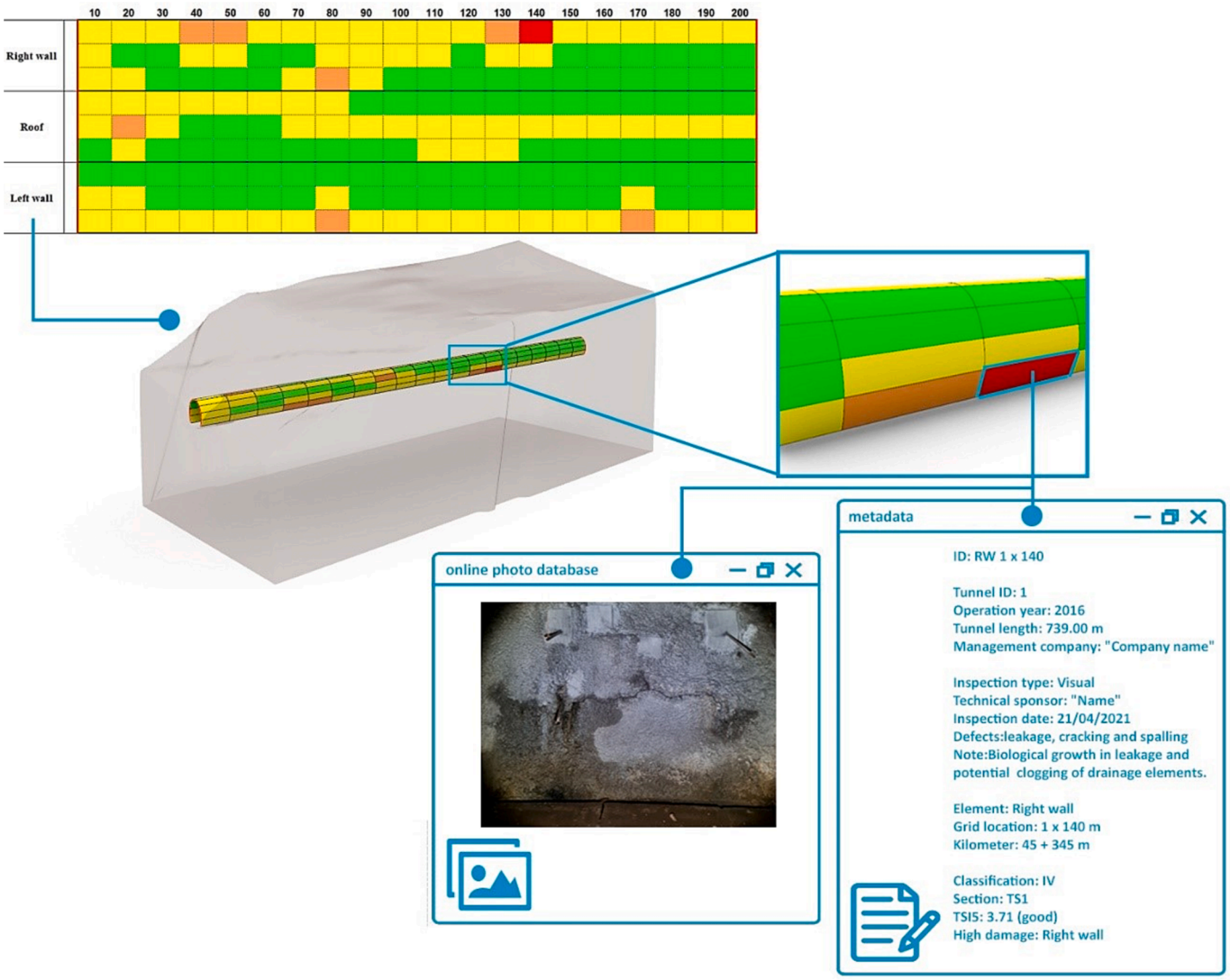

Fig. 20. *IDT* metadata input with EleFront, virtual database and model for *TL 01*.

information during different stages of the tunnel life cycle. Data processing techniques, such as data cleansing, transformation, and integration, are applied to ensure data quality and consistency.

6. Conclusions

The results from the application of the methodology for quantifying service levels were consistent with the initial hypotheses regarding the degradation levels determined from the geological-geotechnical mapping. The analysis also identified areas with the highest rates of deterioration, particularly in the section near the north opening (*TS* 1), which exhibited a service level index (*TSI)* of 3.708. Specifically, the right wall exhibited the lowest *TSI* value of 5.542, indicating a higher level of deterioration in that particular area. These findings provide valuable insights into the areas that require immediate attention and targeted maintenance interventions to ensure the long-term performance and safety of the tunnel.

The methodology employed in this study exhibits flexibility, allowing for further improvements in predictions as more data is collected about the structure. A visual inspection conducted after ten years of tunnel service can provide additional data to calculate the $TSI_{10\ years}$ (service index at ten years). The forecast, based on an inverse exponential equation and the proposed quantification of classified data, is complemented by stochastic methods. The target intervention time, estimated at 16 years by limit analysis, offers the potential to optimize the management of the tunnel performance throughout its life cycle and to enhance monitoring plans. However, further data collection is still required, both for the tunnel under analysis and for other structures, in order to improve the calibration of the predictions and to improve their accuracy.

The construction of the inspection digital twin enables the management and assessment of the tunnel lifecycle using the latest tools from industry 4.0. It leverages the structural knowledge and standardization of the information extracted from the actual asset, attributing it to the virtual model to create a continuous cycle of updates. In future work, efforts will be made to design methods for automating the input of information and the creation of the model using field data, such as point clouds.

Our proposal is to create limited analyses with more detailed service data for a given structure characteristics for future work, to particularize the forecast by simple data, and promote increased confidence. The performance level database for different structures must be implemented. In addition, we also intend to analyze the next steps of the methodology that consist of the regression to calculate α and β in the inverse exponential equation to find the shape of the forecast curve.

CRediT authorship contribution statement

Lucas Bellini Machado: Conceptualization, Methodology, Validation, Writing – original draft, Data curation, Investigation, Visualization. **Marcos Massao Futai:** Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lucas Bellini Machado reports financial support was provided by Coordination of Higher Education Personnel Improvement and Foundation for the Technological Development of Engineering.

Data availability

No data was used for the research described in the article.

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